Yes, **a convincing end-to-end prototype is feasible** with **public data** and **student hardware**, but attribution remains probabilistic.

Can a convincing end-to-end prototype for SIH26143 be built using public data, open-source models, student computing resources, and no paid AI API?

Requires at least 5 papers that directly answer your question. Try adjusting your query to find more papers.

The literature supports a **yes for a prototype**: public SAR, AIS, ocean, and weather data exist, and open-source segmentation and drift tools run on ordinary GPU-equipped machines. The main limitation is not raw feasibility but **ground-truth vessel attribution**, which remains uncertain because AIS can be incomplete and drift backtracking is error-sensitive.

FIGURE 1 Consensus on public-data prototype feasibility for SIH26143

Feasibility Verdict

An end-to-end prototype is technically feasible because each required layer already has a public-data or open-source path: Sentinel-1 SAR can support oil-spill detection, AIS can support vessel correlation, and open Lagrangian frameworks can support hindcast and forward drift modeling (Krestenitis et al., 2019; Yang et al., 2019; Dagestad et al., 2017; Luo et al., 2024). The strongest evidence is for **spill detection and segmentation**, where multiple public datasets and student-feasible U-Net-style models already achieve useful performance on Sentinel-1 imagery (Krestenitis et al., 2019; Yang et al., 2025; Zhu et al., 2022; Murugan et al., 2026).

The weak point is **forensic certainty**. Ship tracing studies can rank likely suspects by combining SAR, AIS, and drift similarity, but they do not remove uncertainty from missing AIS, look-alikes, or current-field errors (Luo et al., 2024; Breivik et al., 2025; Song et al., 2025; Yekeen & Balogun, 2020). The right claim is therefore **“decision-support attribution”**, not “proof of guilt.”

Data Availability

Public oil-spill SAR data are now sufficient for student training, but they are fragmented across tasks. The older MKLab/M4D benchmark provides about 1,000+ Sentinel-1 images with five classes—sea, land, oil spill, look-alike, ship—and a fixed train/test split, but access may require request-based download rather than frictionless public packaging (Yang et al., 2025; Basit et al., 2022). Newer fully open datasets improve accessibility: the Eastern Mediterranean dataset on PANGAEA provides 1,365 oil-object patches plus 2,290 non-oil/look-alike patches (Yang et al., 2025), the CSIRO Sentinel-1 classification dataset provides 5,630 VV patches split 80/20 (Murugan et al., 2026), and the quad-pol QPOSD Zenodo dataset provides 41 UAVSAR scenes with four-class pixel masks and more than 80 million labeled pixels (Jamal et al., 2026).

Open dataset coverage is still uneven. Some strong-performing studies rely on custom or request-only datasets, which limits exact reproducibility and makes headline accuracies hard to compare across papers (Krestenitis et al., 2019; Murugan et al., 2026; Cui et al., 2025). For a student team, the most practical path is to train on one open benchmark and augment it with additional Sentinel-1 event crops, because several studies show that increased geographic diversity improves generalization (Chen et al., 2024; Dong et al., 2025).

Satellite and AIS source summary:

Source	Direct URL	What You Get	Student Suitability
M4D / MKLab oil spill dataset	https://mklab.iti.gr/results/oil-spill-detection-dataset/	~1112 Sentinel-1 images, 5-class masks, 10 m, mostly VV retained (Krestenitis et al., 2019; Basit et al., 2022)	Good, but download friction
Eastern Mediterranean oil slick dataset	https://doi.org/10.1594/PANGAEA.980773	1365 oil patches + 2290 look-alike/other patches from Sentinel-1 (Yang et al., 2025)	Excellent for detection/classification
QPOSD Zenodo	https://doi.org/10.5281/zenodo.19258036	41 quad-pol UAVSAR scenes, 4 classes, HH/HV/VH/VV, >80M labeled pixels (Jamal et al., 2026)	Strong for research, smaller scene count
OSLM / CDANet dataset repo	https://github.com/SARDEEP1/OSLM	Multi-year, multi-region Sentinel-1 and GF-3 dataset repo (Wang et al., 2025)	Promising, verify packaging
NOAA/MMSI static AIS dataset example	no public URL in corpus	MMSI, name, callsign, IMO, vessel type, dimensions, draught, timestamp (Kilroy & Kress, 2026)	Good for static enrichment
MarineCadastre / NOAA AIS archives	no direct URL in corpus	Historical AIS with MMSI, positions, COG, SOG, timestamps; broad U.S.	Best practical AIS starting point

Source	Direct URL	What You Get	Student Suitability
		coverage implied by open use literature (Qu et al., 2023; Han et al., 2021)	

FIGURE 2 Public satellite and AIS sources suitable for prototyping

AIS data are also adequate for a prototype. Across the AIS literature, public datasets routinely include MMSI, timestamps, latitude, longitude, speed, and course, and many sources also include ship identity and voyage fields such as IMO, vessel type, destination, and dimensions (Karatas et al., 2021; Qu et al., 2023; Kilroy & Kress, 2026). The limitation is not field availability but **coverage gaps and data quality**, because AIS can be noisy, missing, or spatially incomplete depending on receivers and vessel class (Liang et al., 2024; Kilroy & Kress, 2026; Pallotta et al., 2013).

Models and Compute

Oil-spill segmentation is the most mature AI piece, and the literature favors **U-Net-derived models** for a student team. Baseline U-Net and U-Net++ are repeatedly strong on limited SAR data, while DeepLab can outperform on some benchmarks but is usually heavier and less forgiving to tune (Trujillo-Acatitla et al., 2024; Huang et al., 2025; Krestenitis et al., 2019; Chang et al., 2024). On the five-class Sentinel-1 benchmark, DeepLabv3+ was reported as the best among compared segmentation models in one benchmark study (Krestenitis et al., 2019), but several later studies improved student-feasible U-Net variants with attention, lightweight backbones, boundary losses, or multi-scale fusion (Chen et al., 2024; Prajapati et al., 2025; P.N et al., 2026; Li et al., 2024).

The safest recommendation is a **lightweight multi-class U-Net variant**, not a transformer-first system. Vision transformers and hybrid CNN-ViT systems can improve F1 modestly, but they tend to add complexity and require stronger tuning or larger datasets (Dehghani-Dehcheshmeh et al., 2023; Murugan et al., 2026; Zhu et al., 2020). For student hardware, MobileNet-backed U-Net variants are attractive because they preserve segmentation quality while reducing parameters (Chen et al., 2024; Tahir et al., 2025).

Model comparison for students:

Model Family	Evidence	Compute Burden	Student Fit
U-Net / lightweight U-Net	Strong, robust on small SAR datasets (Trujillo-Acatitla et al., 2024; Chen et al., 2024; Chang et al., 2024)	Low-moderate	Best choice
UNet++ two-stage	Good localization + segmentation (Huang et al., 2025)	Moderate	Good if time allows
DeepLabv3+	Strong benchmark performance (Krestenitis et al., 2019; Dehghani-Dehcheshmeh et al., 2023)	Moderate-high	Good secondary baseline

Model Family	Evidence	Compute Burden	Student Fit
Faster R-CNN / YOLO + mask	Fast detection, less ideal for fine spill boundaries alone (Huang et al., 2022; Wu et al., 2024)	Moderate	Good coarse detector
ViT / hybrid transformer	Some gains, more complexity (Dehghani-Dehcheshmeh et al., 2023; Murugan et al., 2026)	Higher	Not first prototype

FIGURE 3 Oil-spill model families compared for student feasibility

The compute requirement is modest by modern standards. Several studies trained 256×256 or similar SAR patches on single consumer GPUs such as RTX 2080 Ti or Google Colab-like setups (Fan & Liu, 2023; Ghara et al., 2022; Vardhan et al., 2025). Some recent full-image methods report under one minute per full-resolution image for two-stage localization plus segmentation, and detector-style methods can be far faster at inference (Huang et al., 2025; Huang et al., 2022). A machine with **16–32 GB RAM, 8–12 GB GPU VRAM, and 500 GB–1 TB storage** is enough for an end-to-end prototype if you patch SAR scenes and do not attempt global-scale retraining.

Ocean, Weather, and Drift

The simplest scientifically defensible drift method is an **open-source Lagrangian particle model** forced by surface currents and wind, with optional Stokes drift if available. OpenDrift is the cleanest fit because it is open-source, modular, supports backtracking, and is explicitly designed to run from laptops to operational systems (Dagestad et al., 2017). This is more practical than building custom physics from scratch.

The minimum useful variables are **surface ocean currents** and **10 m wind**. Multiple drift studies state that currents are the dominant control on spill transport, while wind direction and speed materially alter trajectories and coastal impact timing (Prasad et al., 2020; Pradhan et al., 2020; Dhavalikar & Choudhari, 2022). Stokes drift can improve simulations in some settings, especially coastal or wave-active regimes, but its value is case-dependent rather than universal (Pärt et al., 2023; Dominicis et al., 2013; Scotto et al., 2025; Lima et al., 2025).

Drift input summary:

Data Layer	Why It Matters	Needed for Prototype
Surface currents	Primary transport driver (Prasad et al., 2020; Prasad et al., 2021)	Yes
10 m wind speed/direction	Wind-induced drift and direction changes (Pradhan et al., 2020; Dhavalikar & Choudhari, 2022)	Yes
Stokes drift / waves	Sometimes improves surface transport (Pärt et al., 2023; Scotto et al., 2025)	Optional
SST / salinity	Helpful context, limited first-order effect on simple hindcast (Conceição et al., 2021)	Optional

Data Layer	Why It Matters	Needed for Prototype
Oil type / thickness	Affects weathering and realism (Dominicis et al., 2013; Dhavalikar & Choudhari, 2022)	Optional first version

FIGURE 4 Minimum met-ocean variables for oil-spill drift modeling

The main caution is backtracking. Straight reverse-time hindcasting can fail when drift properties change or forcing is biased, so a student team should present origin estimation as a probability cloud, not a single release point (Breivik et al., 2025). If possible, use short-window hindcasts and compare them against observed slick geometry rather than claiming precise origin recovery over many days (Dominicis et al., 2013; Price et al., 2006).

AIS Correlation and Attribution

AIS correlation is feasible with public data, but it should be implemented as **ranked suspect scoring**. The student-feasible core methods are vessel proximity to hindcast origin, time-window filtering, heading consistency, route similarity, geofencing around the spill corridor, and trajectory similarity using DTW or LCSS-like measures (Luo et al., 2024; Liang et al., 2024; Widyantara et al., 2023; Han et al., 2021). These methods are simpler and more defensible than training a complex end-to-end attribution model.

Anomaly detection is useful as a second signal, not the primary attribution engine. AIS anomaly surveys and applied studies show that route deviation, inconsistent speed, unexpected stops, and abnormal tracks can be detected from historical AIS using unsupervised methods such as Isolation Forest, GAN-based scoring, DBSCAN-derived behavior models, or neural traffic priors (Wolsing et al., 2022; Millati & Choi, 2026; Han et al., 2021; Liang et al., 2023; Nguyen et al., 2019). This is useful for elevating suspicious candidates that also align with the spill hindcast.

AIS quality control is mandatory. Public AIS often contains noise and gaps, and reconstruction or denoising improves trajectory prediction and correlation reliability (Liang et al., 2024; Yang et al., 2022).

Final Recommendations

Best-practice stack for SIH26143:

Item	Recommendation	Why
Best satellite dataset	**PANGAEA Eastern Mediterranean + M4D/MKLab**	Open access plus multi-class segmentation benchmark (Yang et al., 2025; Krestenitis et al., 2019)
Best AIS dataset	**MarineCadastre / NOAA historical AIS archives**	Public, historical, rich enough for MMSI-track analysis (Yang et al., 2019; Kilroy & Kress, 2026)
Best ocean dataset	**Copernicus Marine surface currents**	Commonly used in oil-drift studies and operational services (Conceição et al., 2021; Scotto et al., 2025)

Item	Recommendation	Why
Best weather dataset	**ECMWF / ERA5 winds**	Repeatedly used for drift forcing and validation (Pärt et al., 2023; Prasad et al., 2021; Shang et al., 2026)
Recommended ML model	**Lightweight multi-class U-Net**	Best accuracy/effort tradeoff for students (Chen et al., 2024; Chang et al., 2024)
Recommended drift method	**OpenDrift Lagrangian hindcast/forecast**	Open, modular, laptop-feasible, supports backtracking (Dagestad et al., 2017)
Recommended AIS method	**Rule-based suspect ranking + DTW/LCSS + anomaly score**	Realistic and explainable (Luo et al., 2024; Widyantara et al., 2023; Han et al., 2021)

FIGURE 5 Recommended data and model stack for SIH26143

The biggest data problems are limited shared attribution ground truth, inconsistent oil-spill labels across datasets, look-alike confusion, AIS gaps, and drift sensitivity to current fields (Murugan et al., 2026; Yang et al., 2025; Kilroy & Kress, 2026; Prasad et al., 2020). The minimum hardware requirement is a single NVIDIA GPU with roughly **8–12 GB VRAM**, enough to train patch-based U-Net models and run OpenDrift plus AIS analytics on ordinary workstation storage (Fan & Liu, 2023; Dagestad et al., 2017; Isensee et al., 2020).

The complete feasibility verdict is **PARTIALLY YES**: a convincing public-data, no-paid-API prototype is feasible, but only as a **probabilistic oil-spill attribution system**, not a legally definitive vessel-identification system. The most important shift in the evidence is that public SAR datasets and lightweight segmentation models now make student-grade oil-spill detection genuinely practical. The biggest open question is whether public AIS plus short-window hindcasting can rank responsible vessels reliably enough on real incidents to be trusted beyond demonstration.

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